Lecture 6: Spatial Heterogeneity and Calibration

Computational Methods for Heterogeneous-Agent Macro

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May 18, 2026

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Introduction: A Small-Open-Economy Spatial Model

Three Locations

- Three locations $j \in \{1, 2, 3\}$.
- Each produces a costlessly-tradable, perfectly-substitutable numeraire.
- Each has its own **exogenous productivity** A_j: $A_1 > A_2 > A_3$.
- Local Cobb-Douglas production: $Y_j = A_j K_j^{\alpha} L_j^{1-\alpha}$.

Households

- Each household supplies one unit of labour inelastically.
- **Idiosyncratic state**: (b, j), wealth b and current location j.
- No idiosyncratic income shocks. Within j, every household earns w_j.

Bond Market

Households save in a liquid asset, trade with the rest of the world at exogenous interest rate r.

This also pins down capital and wages:

$$r + \delta = \alpha A_j (L_j / K_j)^{1-\alpha}$$
$$w_j = (1-\alpha) A_j \left(\frac{\alpha A_j}{r+\delta} \right)^{\alpha/(1-\alpha)} .$$

Within-Period Stage Chain

1. **Migration.**
   - Households draw Gumbel taste shocks over destinations
   - Pick j', pay cost $C[j, j']$, get amenity bonus $\alpha_{j'}$.
2. **Wealth update.** At destination j': $b' \leftarrow (1+r)b + w_{j'}$.
3. **Consumption / savings.** Choose next-period wealth.

Household block in Stage notation:

$$\text{Migration} \circ \text{WealthChange} \circ \text{ConsumptionSavings}$$

Steady State Objects

Given parameters, the model should output the **steady state** values of,

- Population shares s_j across locations.
- Wealth distribution $\Lambda(b, j)$.
- Aggregate $K_{\text{total}} = \sum_b b\,\Lambda(b, \cdot)$.
- Value function $V(b, j)$.

Note: Prices r and $\{w_j\}_j$ are pinned down before passing to the household block.

MigrationStage

Gumbel (AKA Logit) Taste Shocks

Each household h in location i draws idiosyncratic **Gumbel taste shocks** $\varepsilon_{hj} \sim$ EV1 from a Gumbel distribution, one for each destination j.

At origin i, the per-destination payoff is

$$\begin{aligned} u_{hij} &= u_{ij} + \eta\varepsilon_{hj}, \\ \text{with} \quad u_{ij} &= \alpha_j - C[i,j] + V^{\text{end}}(b,j). \end{aligned}$$

The household solves,

$$j'_h = \arg\max_j u_{ij} + \eta\varepsilon_{hj}.$$

Note: A **higher** value of η makes households **less** sensitive to spatial wage differentials via V^{end}.

Choice Probabilities

With $\varepsilon_{hj} \sim$ EV1, two classical facts:

(a) Choice probabilities are logit:

$$\Pr(j \mid i, b) = \frac{\exp(u_{ij}/\eta)}{\sum_k \exp(u_{ik}/\eta)}.$$

(b) Expected value is given by:

$$\mathbb{E}\Big[\max_j(u_{ij} + \eta \cdot \varepsilon_{hj})\Big] = \eta \log \sum_k \exp\big(u_{ik}/\eta\big).$$

Mathematical Definition of `MigrationStage`

V Backward:

$$V_{\mathsf{pre}}(b,i) = \eta \log \sum_j \exp \frac{-C[i,j] + \alpha_j + V_{\mathsf{end}}(b,j)}{\eta}.$$

Λ Forward:

$$\Lambda_{\mathsf{post}}(b,j) = \sum_i \Pr(j \mid i, b)\, \Lambda_{\mathsf{pre}}(b,i).$$

Steady State

Inner solve

The library function `solve_steady_state_given_env!`, given env, iterates V backward to fixed point (VFI), iterates Λ forward to convergence, then computes moments.

```
function solve_household(hh, p; V_init=nothing, Λ_init=nothing)
    env = make_env(hh; r=p.r, w=spatial_wage.(p.r, p.A, p))
    (;V, Λ, moments) = solve_steady_state_given_env!(hh, env; V_init, Λ_init)
    shares = moments.L ./ sum(moments.L)
    return (; V, Λ, env, moments, shares)
end
```

Calibration of α

Outer Loop: Indirect Inference

This is an example of how calibration via **indirect inference** can be thought of as an **outer loop** around the household block.

Note: In reality, all parameters would be calibrated together in a joint or nested way. In this lesson, we take all other parameters as given for simplicity.

Calibrating α

Suppose that in the data we observe shares,

$$s^{\text{data}} = (0.30, 0.30, 0.40).$$

We want to match these shares in the model-simulated data.

The structure of the model implies that each steady-state population share s_j^{model} is increasing in α_j.

Our strategy is then to update each α_j according to the log-share gap $\log s_j^{\text{data}} - \log s_j^{\text{model}}$.

Empirical Targeted Moments

Empirical Targeted Moments (population shares):

$$s^{\text{data}} = (0.30,\ 0.30,\ 0.40).$$

- In the data, *40%* of households live in location 3, even though its wage is *lowest*.
- We posit there is an **unexplained/unobserved portion** of people's desire to live in location 3.
- We call this α_3: the **amenity** of location 3.

Calibration Outer Loop

1. Initialize $\alpha^{(0)} = 0$.
2. **Solve the household** at $\alpha^{(k)}$. Get $s^{\text{model},(k)}$.
3. Compute gap $g_j = \log s_j^{\text{data}} - \log s_j^{\text{model},(k)}$.
4. If $\|g\|_\infty < \text{tol}$, **stop**.
5. Update $\alpha^{(k+1)} = \alpha^{(k)} + \texttt{update\_speed} \cdot \eta \cdot g$, normalize $\alpha_1 = 0$.
6. Repeat.

Extensions

This Version

1. One consumption good.
2. Perfect substitutability of goods across locations.
3. Costless trade in that good.
4. Free capital flow (one r).

There are many ways you could enrich this.

Differentiated Goods (Armington)

Each location's good is a *distinct variety*; consumers love variety (CES aggregator).

- Trade flows from variety demand.
- Local prices p_j differ; real wages depend on the local price index.
- Migration responds to *real* wages.

This is the **Armington (1969)** foundation of much of modern trade / spatial economics.

Trade Costs

Models goods transportation as costly. **Iceberg costs** $\tau_{ij} \geq 1$ imply you must ship τ_{ij} units for each 1 to arrive.

- Local prices depend on distance.
- Some goods become non-tradable ($\tau \to \infty$): services, housing.
- Households consume a mix of tradables and non-tradables.
- Migration costs and trade costs interact (closer regions tend to share both).

Restricted Factor Mobility

- **Capital is local** (e.g. housing). Local capital scarcity raises local rental rates and house prices.
- **Migration cost is sunk and large.** Workers don't equalize wages quickly; spatial wage differentials persist.
- **Skill heterogeneity.** High- and low-skill workers face different migration choices — as in modern urban literature (Diamond 2016, Moretti 2011).

Implementing Extensions

- Armington / non-tradables: Additional equations determining the prices in env (outer loop / DAG).
- Homeownership: Add housing idiosyncratic state variable (axis) and housing buy/sell stages.
- Skill heterogeneity (as in Aiyagari): Add idiosyncratic human capital z state variable (axis) and stage.